

B.TECH
(SEM VI) THEORY EXAMINATION 2022-23
TRANSPORT PHENOMENON

Time: 3 Hours

Total Marks: 100

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. **2 x 10 = 20**

- a. Differentiate between kinematic viscosity & dynamic viscosity
- b. Write the assumptions of flow through circular pipe?
- c. How is the stream function defined, and why is it useful?
- d. What is time smoothing of equation of change?
- e. What is effect of temperature on thermal conductivity of gases & liquid?
- f. Explain the terms 'steady flow', 'laminar flow'?
- g. Write the assumptions in "diffusion into a falling liquid film"?
- h. Write equation of continuity in rectangular coordinates for incompressible fluid.
- i. Summarize all the steps required in obtaining the equation of change for the temperature
- j. Explain Fick's law of diffusivity.

SECTION B

2. Attempt any three of the following: **10x3=30**

- a. Distinguish between Homogenous and Heterogeneous reactions. Which ones are described by boundary conditions and which ones manifest themselves in the differential equations?
- b. Air at 27°C flows normal to a 73°C, 30 mm o.d. water pipe. The air moves at 1m/s. Estimate the rate of heat transfer per unit length kinematic viscosity=1.624 m²/s, Thermal conductivity=0.0261 W/mk, Npr for air=0.702.
- c. Explain the concept of molecular theory of the viscosity of gases at low density.
- d. Elucidate various boundary conditions encountered in fluid flow.
- e. A substance has a temperature dependent thermal conductivity k (W/m°C) = $0.45 + 8.1 \times 10^{-4} T + bT^2$, where $b = 6.2 \times 10^{-7}$, and T is in °C. What is the unit of the coefficient b ? Find the thermal conductivity of the material at 1050°F.

SECTION C

3. Attempt any one part of the following: **10x1=10**

- a. What is diffusion? Explain the temperature & pressure dependence of mass diffusivity?
- b. Obtain an expression for mass rate of flow w for an ideal gas in laminar flow in a long circular tube. The flow is presumed to be isothermal. Assume that the pressure change through the tube is not very large, so that the viscosity can be constant.

4. Attempt any *one* part of the following: 10x1=10

- a. Write the comparison between forced and free convection in a non-isothermal system?
- b. What is a composite wall? Derive the relevant equations for heat conduction through composite wall?

5. Attempt any *one* part of the following: 10x1=10

- a. Derive an expression for temperature distribution $T(x)$ in a viscous fluid in steady laminar flow between large flat parallel plates kept $2B$ apart. Both plates are maintained at constant temperature T_0 . Take in to account explicitly the heat generated by viscous heat dissipation. Neglect the temperature dependence of μ and k .
- b. Calculate the thermal conductivity of a gas mixture containing 20 mol% CO_2 and 80 mol% H_2 at 1 atm and 300K. Given Thermal Conductivity of CO_2 and H_2 are 383×10^{-7} (Cal/s.cm.K) and 4250×10^{-7} (Cal/s.cm.K) respectively & Viscosity of CO_2 is 1495×10^{-7} (g/s.cm) and H_2 is 896×10^{-7} (g/s.cm).

6. Attempt any *one* part of the following: 10x1=10

- a. Using Navier-Stokes equation, derive an expression for (i) Velocity distribution; (ii) Maximum velocity; and (iii) Average velocity for the flow of an incompressible Newtonian fluid flows vertically downward through circular tube of radius R at steady state.
- b. Explain partial, substantial and total derivative.

7. Attempt any *one* part of the following: 10x1=10

- a. What are the various types of fluxes involved in mass transfer.
- b. Calculate steady state mass flux of helium at 500°C . The partial pressure of helium is 1atm at lower end and zero at the upper surface of the plate. Thickness of the pyrex plate is 10^{-2} mm and its density is 2.6 g/cm^3 . The solubility and diffusivity of helium in pyrex are reported as 0.0084 volume of gaseous helium per volume of glass and $0.2 \times 10^{-7} \text{ cm}^2/\text{s}$ respectively.